Jio Oh

Seoul, Republic of Korea | harryoh99@kaist.ac.kr | www.jiooh.com

HIGHLIGHTS

I am a second-year PhD student in Data Intelligence Lab at KAIST, advised by Prof. Steven E. Whang. Most recently, I was a applied scientist intern in Amazon Alexa International Team in Bellevue, WA, advised by Paul Vicinanza, Dezhi Hong, and Amani Namboori. I was also previously a research intern in Microsoft Research Asia (MSRA), VUNO, and NCSoft. I am interested in generative models and large language model (LLM) agent personalization, improving and evaluating the trustworthiness of LLMs by taking data-centric approaches, and in applications of LLMs in social science.

Education

KAIST, Daejeon, Korea, PhD in Electronic Engineering — Mar 2025 -

- Data Intelligence Lab (Advisor: Steven Euijong Whang)

KAIST, Daejeon, Korea, MS in Electronic Engineering — Mar 2023 - Feb 2025

- Data Intelligence Lab (Advisor: Steven Euijong Whang)
- GPA: 4.05/4.3

KAIST, Daejeon, Korea, BS in School of Computing — Mar 2018 - Feb 2023

- Double Major: Industrial and Systems Engineering
- GPA: 3.91/4.3 (Major: 4.01/4.3)

Georgia Institute of Technology, Atlanta, GA, USA — Jan 2022 - May 2022

- Exchange Student (Major: Computer Science, College of Computing)

Experience

Research Assistant, Data Intelligence Lab, KAIST, Dajeon, Korea — Mar 2023 -

- Research Interests: LLM, GenAI Personalization, LLM Agents, and Data-Centric AI
- Showed that LLMs show better performance and robustness when data are represented in tabular formats compared to other semi-structure or text formats. (***Under Review***)
  - Supported by MSRA and Microsoft Accelerate Foundation Model Research Program
- Proposed an LLM evaluation benchmark that is automatically generated from relational databases and can verify LLM reasoning by utilizing database integrity constraints (***NeurIPS'24 Spotlight***)
  - Responsible for ideation and experiments
  - Supported by MSRA and Microsoft Accelerate Foundation Model Research Program

Applied Scientist Intern, Amazon, Bellevue, WA — Aug 2025 - Dec 2025

- Worked for Alexa International Team, Alexa, focusing on research on multi-dialectal aspects of LLMs.
- Proposed a novel dialect-aware data generation framework spanning various pillars of written dialect. (***Under Review***)
  - Close interactions with nine different country team members (research, project management, customer support) and linguists.
  - Focused on generation of a large-scale dataset, LLM evaluation, and post–training.
- Mentor: Paul Vicinanza, Dezhi Hong, and Amani Namboori

Research Intern, Microsoft Research Asia, Beijing, China — Sep 2024 - Mar 2025

- Worked for Social Computing Group, focusing on research on LLM for education.
- Showed that LLMs show better performance and robustness when data are represented in tabular formats compared to other semi-structure or text formats. (***Under Review***)
  - Supported by MSRA and Microsoft Accelerate Foundation Model Research Program

- Proposed a grade-specific data generation and LLM fine-tuning framework that adapts educational explanations to students' comprehension levels while preserving factual accuracy. (***npj Artificial Intelligence 2026***)
- Mentor: Jindong Wang and Xing Xie

Research Intern, VUNO, Seoul, Korea Jul 2022 - Aug 2022

- Worked for Brain Lung 3D Team focusing on brain medical AI
- Developed an unprecendented multi-modal deep learning model for ICV Segmentation with both T1 and T2 brain MRI images (***Frontiers in Neurology 2023***)
- Mentors: Wooseok Jung and Jinyoung Kim

UG Research Assistant, Computational Data Science Lab, GATECH, Atlanta, GA Jan 2022 - Jul 2022

- Worked on a collaborative project with Wikipedia deploying a website that can be used for detecting ban evasion from malicious users
- Presented the demo tool in the Wiki Workshop at The Web Conference 2022
- Mentors: Gaurav Verma and Srijan Kumar

R&D Intern, NCSoft, Pangyo, Korea Jan 2021 - Feb 2021

- Worked for Financial Intelligence Team, Knowledge AI Lab, Natural Language Processing Center
- Researched whether deep learning models can predict the next year's growth of firms based on attributes from financial statements and macroeconomic variables
- Mentors: : Heehwan Park and Jung Han

Research Intern, Interactive Computing Lab, KAIST, Daejeon, Korea Jul 2020 - Dec 2020

- Researched and experimented how human behavior changes with tolerance on goal setting introducing a new concept, "margin" in terms of goal setting (***CHI' 21***)
- Mentors: : Gyuwon Jung and Uichin Lee

HONORS

KAIST Best TA Award, KAIST EE Department Spring 2025
KAIST Dean's List, KAIST College of Engineering Fall 2021
KAIST Dean's List, KAIST College of Engineering Spring 2021
KAIST IE Frontier, 3rd place, KAIST ISYSE Department 2020
KAIST Engineering Innovator Award, KAIST College of Engineering 2020

Publications

J. Oh, P. Vicinanza, T. Butler, S. E. Whang, D. Hong, and A. Namboori. "DialectLLM: A Dialect-Aware Dialog[ue] Generation Framework Beyond Standard American English". *Under Review*, 2026.

J. Oh, S. E. Whang, J. Evans, and J. Wang. "Classroom AI: Large Language Models as Grade-Specific Teachers". In ***npj Artificial Intelligence*** 2(1):28, 2026.

J. Oh, G. Heo, S. Oh, J. Wang, X. Xie, and S. E. Whang. "Talking with Tables for Better LLM Factual Data Interactions". *Under Review*, 2024.

J. Oh, S. Kim, J. Seo, J. Wang, R. Xu, X. Xie, and S. E. Whang. "ERBench: An Entity-Relationship based Automatically Verifiable Hallucination Benchmark for Large Language Models". In *38th Anuual Conference on Neural Information Processing Systems (**Spotlight, NeurIPS'24**), Vancouver, Canada*, 2024.

P. S. Suh, W. Jung, C. H. Suh, J. Kim, **J. Oh**, H. Heo, W. H. Shim, J. Lim, J. Lee, H. S. Kim, and S. J. Kim. "Corrigendum: Development and validation of a deep learning based automatic segmentation model for assessing intracranial volume: comparison with NeuroQuant, FreeSurfer, and SynthSeg". In ***Frontiers in Neurology***, 2023.

G. Jung, **J. Oh**, Y. Jung, J. Sun, H. K. Kong, and U. Lee. "Good Enough!: Flexible Goal Achievement with Margin-based Outcome Evaluation". In *CHI Conference on Human Factors in Computing Systems (**CHI' 21**), Yokohama, Japan*, 2021.

Talks

William&Mary DATA440/669 Trustworthy AI Guest Lecture, Apr 2025
Hallucination Benchmarking for LLMs with Relational Databases

Microsoft Research Asia Lightning Talk, ERBENCH Nov 2024

Microsoft Research Asia Intern Tech Talk, ERBENCH Nov 2024

Professional Services

Program Committee, Invited Reviewer

- NeurIPS, 2026
- NeurIPS, 2025
- IEEE BigComp, 2025
- ACM TIST, 2024

Extracurricular Activities / Leadership

KAIST Financial Analysis Club, KAIST, Daejeon, Korea Sep 2022 - Jun 2023
- Led the club as a club executive (team leader)

Aquila, KAIST, Daejeon, Korea Mar 2019 - Dec 2021
- Led the soccer & futsal club as a club executive

KAIST Buddy Program, KAIST, Daejeon, Korea Spring 2021, Fall 2022
- Helped international students adapt to Korea & KAIST

KAIST Futsal Tournament Champions, KAIST, Daejeon, Korea Spring 2025

KAIST EE Futsal Tournament Champions, (Captain) KAIST, Daejeon, Korea Fall 2023

KAIST Futsal Tournament Champions, KAIST, Daejeon, Korea Spring 2023

Languages

Korean (Native), **English** (Fluent, **TOEFL 118**), and **Chinese** (Basic)